



How Objects Move in Different Ways

Exploring Roll, Bounce and Movement

Learning about Size, Shape and Material



Objects move in different ways! Let's explore why.

Unit Overview

In this unit, students explore how different objects move. They investigate how shape, size and material affect the way objects roll, bounce, fly, spin and stop.

What We Want Students to Understand

- ✓ Objects move in different ways.
- ✓ The shape, size and material can change how an object moves.
- ✓ We can observe, compare and describe movement.
- ✓ First Nations Australians have used traditional toys to teach skills, stories and knowledge.

1. Objects Can Move

Objects can move in many different ways.

Roll	Bounce	Slide	Fly / Throw	Spin
Moves along the ground.	Goes up and down.	Moves across a surface.	Moves through the air.	Turns around and around.

2. What Affects How Objects Move?

Three main things affect how an object moves.

Shape	Size	Material
The shape can help or change the movement.	The size can make it move faster, slower, farther or shorter.	The material can affect how it rolls, bounces, spins or makes sound.

3. How Shape and Size Change Movement

Example	What We Notice	How It Moves	Why? (Shape and Size)
Blocks, tubes or eggs rolled down a slope 	They roll in different ways.	Some go straight. Some wobble or roll in curves.	Curved or round shapes roll. Flat shapes slide or may not roll. Unusual shapes wobble.
Tennis ball, golf ball, marble, basketball (comparing size) 	Different sizes affect how far they roll and how high they bounce.	Larger balls may roll slower but bounce higher. Smaller balls may roll faster.	Bigger size = more weight (may roll slower) but more bounce height. Smaller size = less bounce and may travel less far.
Balls made from different materials 	The same shape can move differently.	Some roll fast, some slow. Some bounce high, some low.	Hard and smooth materials roll fast and bounce high (e.g. plastic, rubber). Soft or rough materials roll slow and bounce low (e.g. foam, cloth).

4. Traditional Instructive Toys of First Nations Australians

These toys are used to teach skills, stories and knowledge while having fun!

Boomerang	Kylies (Dice)	Woomera	Bullroarer
Curved shape thrown through the air. It returns or curves back.	Small and shaped to land and roll in different ways. They tumble and roll on the ground.	Long and curved to help a spear travel farther. It helps the spear travel straight and far.	Flat and balanced to spin and make a deep sound. It spins smoothly in the air.

First Nations Knowledge

Traditional instructive toys are cleverly designed. The shape, size and material help them move in special ways. These toys are an important part of culture and Country.



Always acknowledge the Traditional Owners of the land and their knowledge.

5. Let's Investigate!

We can explore and observe.

- Roll different objects down a slope.
- Bounce balls of different sizes and materials.
- Compare how far they go.
- Talk about what you notice.



6. Key Vocabulary

roll – to move by turning over on a surface
bounce – to go up and come back down
slope – a slanted surface
shape – the form or outline of something
size – how big or small something is
material – what something is made from
movement – how something moves or changes position
spin – to turn around quickly
balance – to stay steady or in control
distance – how far something travels

7. What We Learned

- ★ Objects move in different ways.
- ★ Shape, size and material affect how they move.
- ★ We can observe, compare and describe how objects move.
- ★ Traditional First Nations toys show clever design and teach important skills and knowledge.



Observing helps us understand the world!

Teaching Tips

Use real objects and hands-on activities. Encourage students to make predictions, observe carefully and explain their thinking using simple language.

Cross-Curriculum Links

Maths – measurements, size comparisons, pattern

HASS – cultures, First Nations histories and traditions

Assessment Ideas

- Observation checklists
- Photos or drawings of investigations
- Simple explanations in words